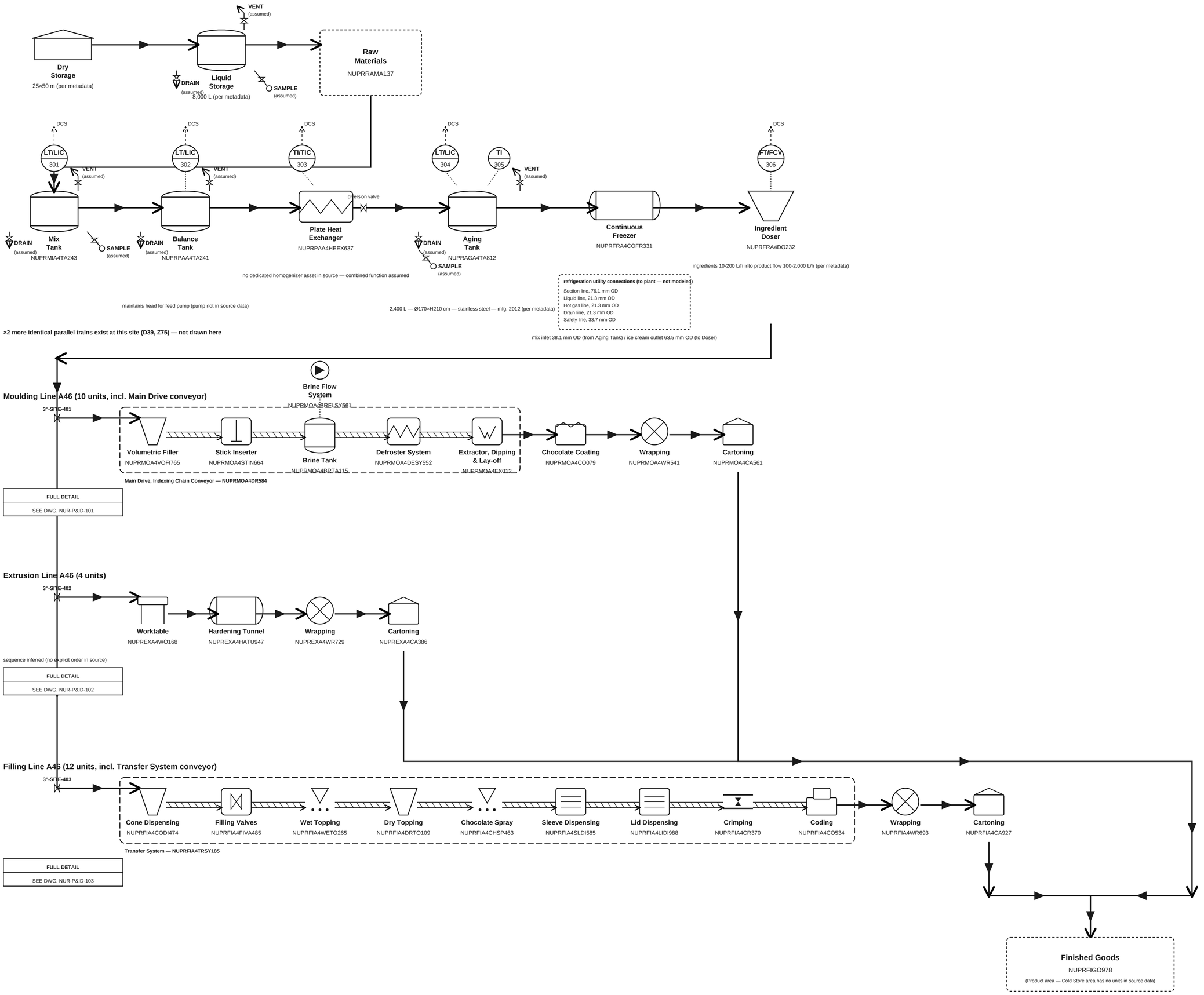
P&ID (PRELIMINARY) — NUREMBERG SITE OVERVIEW — STORAGE THROUGH PACKAGING — ICE CREAM FACTORY DATA

Cognite Data Fusion Core Data Model | ISA-5.1 instrument tagging | For L100 practitioner training use only — not for construction



Legend

Solid line + ►	Process piping / product flow (arrowheads = assumed direction)
Bowtie (on a line)	Manual isolation valve
size"-SITE-4##	Line number: size-service code-sequence (illustrative)
Short dashed line	Instrument signal to PLC / DCS
Dashed box (fine)	Logical/tracking node (Raw Materials, Finished Goods) — not physical equipment
Dashed box (coarse)	Utility connection point not otherwise modeled (e.g. refrigeration plant)
Zigzag-filled box	Plate heat exchanger
House-shaped box	Warehouse / dry storage zone
Circle + triangle	Pump (Brine Flow System)
Circle + cross	Rotary equipment (Wrapping)
Box + open flaps	Cartoning
Downward funnel	Hopper / dispensing (Filler, Cone/Dry Topping dispensing)
Triangle + dots	Spray/liquid application (Wet Topping, Chocolate Spray)
Box + bowtie	Valve bank (Filling Valves)
Box + rod	Linear inserter/plunger (Stick Inserter)
Box + pincer mark	Extraction/gripper mechanism (Extractor, Dipping & Lay-off)
Box + wavy line	Dip/bath coating (Chocolate Coating)
Box + horiz. lines	Stacked-sheet dispenser (Sleeve/Lid Dispensing)
Box + opposing arrows	Press/crimp mechanism (Crimping)
Box + small box	Stamp/coder (Coding)
Box + legs	Worktable
LT / LIC	Level transmitter / level indicating controller (tanks)
TI / TIC	Temperature indicator / temperature indicating controller
FT / FCV	Flow transmitter / flow control valve (dosing, throughput)

Notes / open items before this drawing is construction-ready

This is ONE of three identical parallel trains at Nuremberg (A46 shown; D39 and Z75 are complete duplicates — see all_sites_and_assets CSV). Drawing all three would be pure repetition.

Every asset in all three product lines is now shown individually. All three now have dedicated P&IDs for full instrument-level detail (bubbles, valves, line numbers per unit) — this sheet shows what exists and in what order, not full loops.

Drawing register: NUR-P&ID-000 (this sheet), NUR-P&ID-101 (Moulding Line A46), NUR-P&ID-102 (Extrusion Line A46), NUR-P&ID-103 (Filling Line A46).

Aging Tank capacity/dimensions and Continuous Freezer piping diameters ARE real values from source metadata — everything else (order, groupings, valves, line numbers) is inferred using the same engineering-assumption approach as the per-line P&IDs.

No dedicated homogenizer asset exists in the source data even though the process cell is named 'Pasteurization and Homogenization' — the Plate Heat Exchanger is assumed to cover both functions, unlike a real dairy plant which would split them.

No pump assets exist anywhere in the source data (timing pump, feed pump, etc. are implied by process logic, not modeled).

Cold Store (area) has zero units in the source data — finished product cold-storage is not represented at unit level.

planned_status (Core Data Model time series) is a software/planning variable only — intentionally not drawn on this P&ID.

Source: Ice Cream Factory data, Nuremberg site, A46 train, modeled via the Core Data Model (ProductionLine / Unit / Equipment).

Drawing No.	NUR-P&ID-000
Client / project	Ice Cream Factory — L100 practitioner training
Site	Nuremberg
Scope	Storage -> Packaging, A46 train (1 of 3 parallel trains)
Drawing type	P&ID — preliminary, mixed detail (ISA-5.1)
Data source	Core Data Model (ProductionLine, Unit, Equipment)
Status	For training use — not for construction
Rev.	0 — draft